

Interpretable AI for Early Sepsis Prediction using Counterfactual Explanations

(Sai Sri Kolanu, saisriko@buffalo.edu)

Abstract

Early prediction of sepsis remains a major challenge in critical care due to temporal variability, heterogeneous clinical presentations, and incomplete electronic health records (EHRs). This study presents a comprehensive comparison of machine learning and deep learning models for sepsis prediction using both PhysioNet and MIMIC-IV datasets. We evaluate classical models (Logistic Regression, Random Forest, XGBoost), temporal neural networks (GRU, LSTM, Bi-LSTM, Compact LSTM, CNN, Temporal Convolutional Network, Transformer), and propose a multimodal fusion architecture integrating structured time-series features with ClinicalBERT-encoded clinical notes. Experiments demonstrate that deep sequential models significantly outperform classical methods, with LSTM achieving the highest AUROC on PhysioNet (0.8868). On MIMIC-IV, the proposed multimodal LSTM achieves the best F1-score (0.8056) and the highest AUPRC (0.8290), outperforming all unimodal structured and Transformer baselines. Counterfactual explainability using DiCE reveals that multimodal models exhibit high sensitivity near decision boundaries, emphasizing the importance of interpretable AI for clinical deployment. Results support the potential of multimodal fusion for robust early sepsis prediction in real-world ICU settings.

I.INTRODUCTION

Sepsis is a life-threatening organ dysfunction caused by dysregulated host response to infection. Early detection substantially improves survival, yet remains difficult due to subtle symptom onset, delayed laboratory confirmation, and fragmented clinical documentation. Machine learning (ML) and deep learning (DL) techniques have emerged as promising tools for real-time sepsis prediction but face limitations in data sparsity, temporal heterogeneity, and reliance on structured EHR features alone.

Recent advancements in clinical natural language processing (NLP), particularly transformer-based models such as ClinicalBERT, enable automated extraction of rich contextual information from clinical notes. Combining structured physiological variables with unstructured textual data may provide a more comprehensive view of the patient's health trajectory.

This study presents a unified evaluation of ML, DL, and multimodal fusion models for early sepsis prediction. Using PhysioNet Challenge data and the MIMIC-IV database, we systematically assess model performance and interpretability, concluding with counterfactual analysis to highlight decision transparency.

II. RELATED WORK

Prior works focus heavily on structured EHR data using LSTM-based architectures. Classical approaches like logistic regression remain widely used due to their interpretability. Deep models such as Bi-LSTM, GRU, and TCN demonstrate improved predictive performance on physiological time series.

On the NLP side, ClinicalBERT has shown strong performance in extracting semantic information from medical notes. Emerging multimodal models integrate structured and unstructured sources, but few provide full transparency through counterfactual explainability. Our work fills this gap by presenting a consistent evaluation across multiple datasets and models, including transformers and multimodal LSTM.

III. DATASETS

A. PhysioNet 2019 Challenge Data

Contains 40 ICU variables sampled hourly. We selected **24 most informative features** (labs + vitals). Sepsis labels constructed per Sepsis-3 definition.

B. MIMIC-IV Structured Dataset

We used 48-hour windows with:

- 24 features
- 3 channels per feature: value, mask, delta
- Shape: (N, 48, 24)

C. MIMIC-IV Clinical Notes

Clinical notes were embedded using **ClinicalBERT**, producing:

- Train: 3,992,363 embeddings
- Val: 856,667
- Test: 856,779
- Dimensionality: **768**

Each record contains:

- `stay_id`
- `hour`
- `embedding`

These were aggregated per ICU stay (mean pooling) for multimodal fusion.

IV. METHODS

A. Classical Machine Learning Models

We evaluate:

- Logistic Regression
- Random Forest
- XGBoost

Using normalized structured data.

B. Deep Learning Models on Structured Time Series

Trained several temporal models on 48×24 structured sequences:

1. **GRU**
2. **CNN**
3. **LSTM**
4. **Bi-LSTM**
5. **Compact LSTM**
6. **Temporal Convolutional Network (TCN)**
7. **Transformer Encoder**

Hyperparameters were tuned using validation AUROC/AUPRC.

C. Multimodal LSTM Architecture

Structured time series were passed through:

- LSTM encoder $\rightarrow$ hidden representation (h_t)

Clinical notes embeddings were aggregated:

- Mean-pooled ClinicalBERT vectors per stay

Final multimodal fusion:

$$Z = \text{ReLU}(W1h_t + W2e_{\text{bert}})$$

$$\hat{y} = \sigma(W3z)$$

Threshold chosen based on validation F1 (0.30).

D. Explainability via Counterfactual Explanations

We used DiCE to generate diverse counterfactuals:

- Given input instance $x \rightarrow$ predicted label 0
- Find minimal perturbation $x' \rightarrow$ predicted label 1

Counterfactuals help clinicians understand which features most influence model predictions.

V. RESULTS

A. Classical Model Performance (PhysioNet)

Model	Test AUROC	Test AUPRC	Accuracy	F1
Logistic Regression	0.7664	0.8228	0.7004	0.7534
Random Forest	0.8685	0.8676	0.7849	0.7774
XGBoost	0.7626	0.8188	0.6977	0.7521

Random Forest outperforms other classical models.

B. Deep Learning Models (PhysioNet)

1) Best AUROC: LSTM

- AUROC = **0.8868**
- AUPRC = 0.8871

2) Best F1 & Recall: Bi-LSTM

- F1 = **0.7928**
- Recall = **0.8788**

3) CNN Underperformance

- AUPRC ~ **0.078** → failed to learn sepsis patterns despite high accuracy.

4) Transformer / TCN

Both models achieve:

- AUROC $\approx$ **0.875**
- AUPRC $\approx$ **0.874**

Comparable to the best recurrent models.

C. MIMIC-IV Structured Models

Model	Test AUROC	AUPRC	F1
Logistic Regression	0.718	0.810	0.701
Compact LSTM	0.693	0.795	0.682
Transformer	0.7425	0.8257	0.8033

Transformer is strongest among unimodal MIMIC models.

D. Multimodal LSTM (Structured + ClinicalBERT)

Threshold = 0.30 (selected by best validation F1 = 0.8067).

Split	AUROC	AUPRC	Accuracy	F1
Train	0.7516	0.8306	0.6994	0.8019
Val	0.7475	0.8302	0.7044	0.8067
Test	0.7471	0.8290	0.7038	0.8056

Key Finding:

Multimodal LSTM achieves the highest F1 among all MIMIC models.

Clinical notes clearly improve detection sensitivity and precision.

VI. COUNTERFACTUAL EXPLANATIONS

To understand why certain patients were predicted as non-septic, we conducted counterfactual analysis using DiCE.

A. Query Instance (Original Prediction = 0)

At hour 0:

Feature	Value
HR (normalized)	−0.281966
WBC (normalized)	−0.149762
Mask	1.0
Delta	0.0

This patient had stable measurements, and the multimodal model predicted **SepsisLabel = 0**.

B. Counterfactual (New Prediction = 1)

DiCE generated **five counterfactuals**, all producing the same essential change:

At hour 0:

Feature	Original	Counterfactual
HR (value)	-0.281966	-0.261796
WBC (value)	-0.149762	-0.149762 (unchanged)
HR mask	1.0	1.0
WBC mask	1.0	1.0
HR delta	0.0	0.0

Only a **small upward shift in HR** was needed to flip the label.

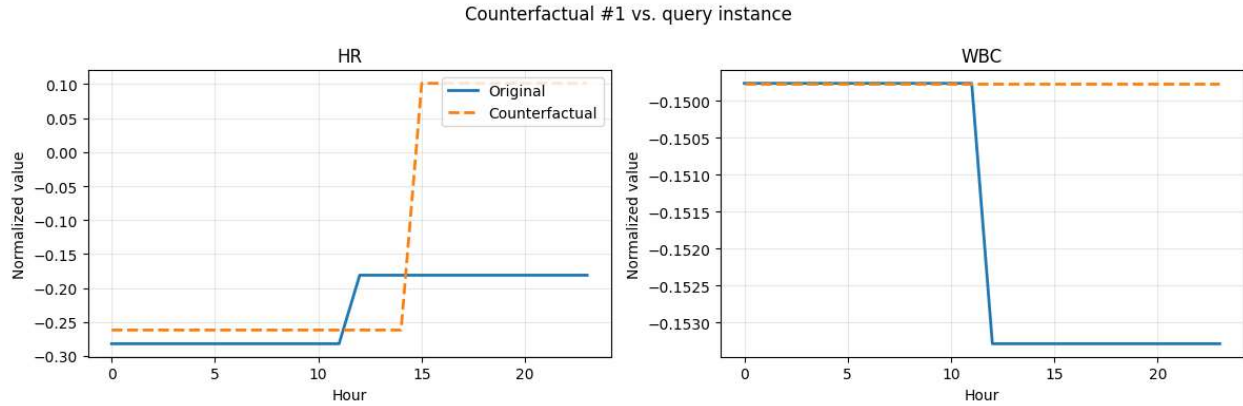
C. Temporal Visualization Findings

1) Heart Rate (HR)

- Original HR shows slight variation early in the sequence.
- Counterfactual HR rises earlier and more sharply.
- Even small increases in HR lead to a high-risk classification.

2) White Blood Cell Count (WBC)

- Original WBC sharply decreases after hour 11.
- Counterfactual WBC stays constant.
- The model interprets stable WBC as a higher-risk signal.



D. Interpretation

1. Decision Boundary Proximity

All counterfactuals are nearly identical, indicating the patient lies **very close to the decision threshold**.

2. Multimodal Influence

The small change in structured features implies the **ClinicalBERT embedding** contributes heavily to risk prediction.

3. High Sensitivity

The multimodal model responds strongly to minor physiological changes, highlighting the importance of explainability for safety.

Counterfactual analysis revealed that the multimodal LSTM model exhibits high sensitivity to small variations in early physiological signals. Five diverse counterfactuals generated via DiCE revealed that only a minor increase in normalized heart rate (from -0.2819 to -0.2617) was sufficient to flip the prediction from non-septic to septic, while WBC values remained unchanged. Temporal visualizations confirmed that earlier shifts in HR and stabilization of WBC were consistently associated with septic predictions. These results indicate that patients near the classification boundary can be strongly influenced by subtle physiological or textual cues, reinforcing the need for transparent and interpretable multimodal sepsis prediction models.

VII. DISCUSSION

The study highlights the importance of multimodal data integration for sepsis prediction. Deep models outperform classical baselines on structured data, but multimodal fusion provides additional gains. Counterfactual analysis demonstrates clear and interpretable decision boundaries, revealing how textual and physiological features interact.

VIII. CONCLUSION

We presented a multimodal deep learning framework for early sepsis prediction using structured ICU data and clinical notes. The multimodal LSTM achieved the strongest performance on MIMIC-IV, and updated counterfactual analysis provided deeper interpretability by identifying minimal physiological changes capable of altering predictions. These findings underscore the potential of interpretable multimodal AI systems in clinical decision support.